Anatomy of One Tree in the Gradient Boosting Model

BME 5650: Biosolid Mechanics, HW 7 | Model from `gbr_boost.ipynb` / `gbr_cv20.ipynb` (same split and settings): 20 trees, learning rate ν = 0.2, max_depth = 2, min_samples_leaf = 10, subsample = 0.7; test RMSE 4.09 GPa, R^2 0.740

1. Where a tree fits in the model

The gradient boosting model is a sum of small regression trees added to a constant starting guess:

$$E(Ca, V_f) = F_0 + \nu \cdot [h_1(Ca, V_f) + h_2(Ca, V_f) + \ldots + h_{20}(Ca, V_f)]$$

F_0 = 16.86 GPa is the mean modulus of the 277 training specimens. Each tree h_m is fit to the *residuals* (measured E − current prediction) left by everything before it, and its output is multiplied by ν = 0.2 before being added. Because subsample = 0.7, each tree is grown from a random 193 of the 277 training specimens. This document takes the first tree, h_1, apart: what each box means, how its splits and leaf values were calculated, and how it changes a prediction.

2. The diagram

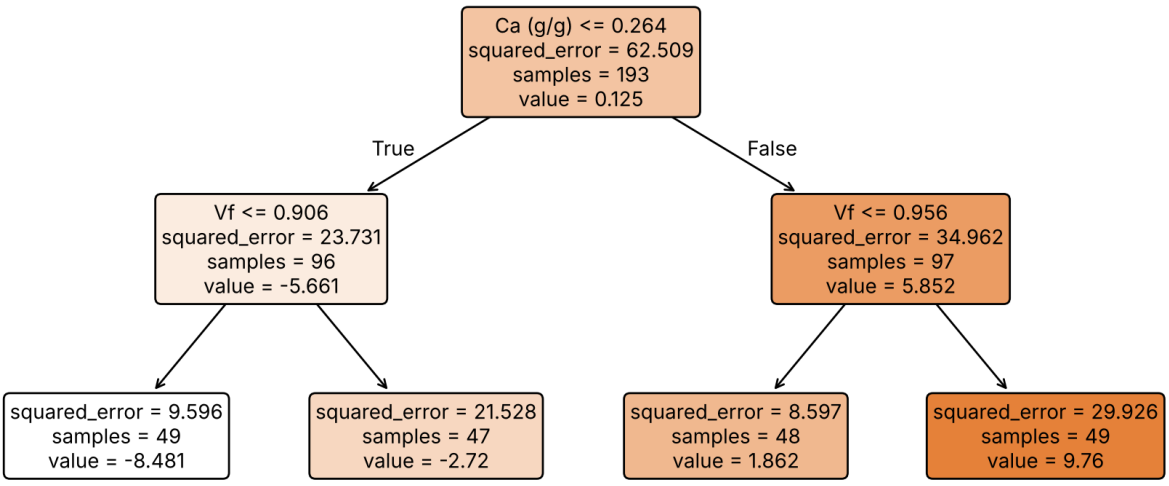


Figure 1. Tree h_1 as drawn by sklearn.tree.plot_tree. Boxes with a rule are split nodes; the bottom row are leaves. Following 'True' (left) means the rule holds; 'False' (right) means it does not. Darker orange = larger value.

Line in a box	Meaning
Ca (g/g) <= 0.264	The split rule (split nodes only). Specimens with Ca ≤ 0.264 g/g (264 mg/g) go left, the rest go right.
squared_error = 62.509	Spread of the residuals in this node: their mean squared deviation from the node's mean (GPa^2). Splits are chosen to make this drop.
samples = 193	How many of the 193 specimens drawn for this tree reach the node.
value = 0.125	Mean residual of those specimens (GPa). In a leaf this is the leaf value, the correction the tree outputs.

Table 1. How to read each box.

3. The same tree as regions of the (Ca, Vf) plane

A depth-2 tree asks two yes/no questions, so it cuts the composition plane into at most four rectangles, one per leaf. Every specimen in a rectangle receives the same correction.

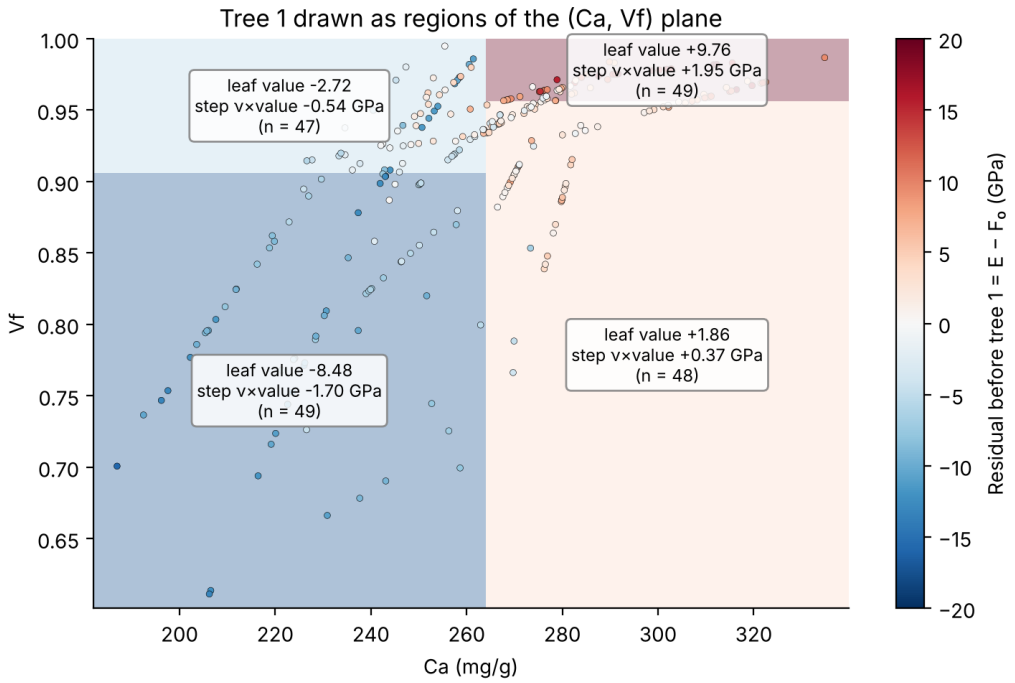


Figure 2. Tree h_1 as a map. Dots are all 277 training specimens, coloured by their residual before tree 1 ($E - F_0$): blue = softer than the mean, red = stiffer. Rectangle shading = leaf value. n counts only the specimens drawn for this tree.

Leaf (region)	Specimens	Spread (GPa²)	Leaf value (GPa)	Step added, ν × value (GPa)
Ca (g/g) ≤ 0.264, Vf ≤ 0.906	49	9.60	-8.481	-1.696
Ca (g/g) ≤ 0.264, Vf > 0.906	47	21.53	-2.720	-0.544
Ca (g/g) > 0.264, Vf ≤ 0.956	48	8.60	+1.862	+0.372
Ca (g/g) > 0.264, Vf > 0.956	49	29.93	+9.760	+1.952

Table 2. The four leaves. Low-Ca, low-Vf bone is about 8.5 GPa softer than the mean; high-Ca, high-Vf bone is about 9.8 GPa stiffer. The first tree has already learned that E rises with both inputs, and that the Vf threshold that matters depends on Ca (0.906 vs 0.956), which is an interaction.

4. How the splits were chosen

Starting from the top node, the tree tries every input and every threshold halfway between neighbouring sorted values, and keeps the split that reduces the spread of the residuals the most. For the root split **Ca ≤ 0.264**:

	Node	Specimens	Spread of residuals (GPa²)
Before	root	193	62.51
After	left: Ca ≤ 0.264	96	23.73
	right: Ca > 0.264	97	34.96
	weighted average of the two	193	29.38

The split cuts the spread from 62.5 to 29.4 GPa², a reduction of 33.1 GPa² (more than half). No other Ca or Vf threshold did better, so it became the root. Each child is then split the same way, using only its own specimens: Vf ≤ 0.906 on the low-Ca side and Vf ≤ 0.956 on the high-Ca side. Growth stops at depth 2. Any split that left fewer than 10 specimens in a leaf would not have been allowed. (sklearn's default criterion, `friedman_mse`, ranks splits by an equivalent score, $n_L n_R/(n_L+n_R) \cdot (\text{mean}_L - \text{mean}_R)^2$.)

5. How the leaf values were calculated

Each leaf value is the **average residual of the specimens in that leaf**, i.e. mean measured E in the leaf − mean current prediction. Before tree 1 the current prediction is F_0 for everyone, so for the bottom-left leaf the value −8.48 GPa means those 49 specimens average 8.48 GPa below the 16.86 GPa mean. The average is used because it is the single number that minimises the squared error within the leaf. The root's value (0.125, not 0) is the mean residual of the 193 drawn specimens, which differs slightly from the mean of all 277.

The tree does not apply the full correction. Each prediction moves by ν × leaf value, one-fifth of the way (ν = 0.2), so the bottom-left specimens move from 16.86 to 16.86 − 1.70 = 15.16 GPa, and the remaining gap is left for later trees.

6. Following one specimen through the model

Test specimen 1: Ca = 0.2559 g/g (256 mg/g), Vf = 0.9654, measured E = 20.83 GPa.

1. Root: is Ca ≤ 0.264? 0.2559 ≤ 0.264 → **yes**, go left.
2. Next node: is Vf ≤ 0.906? 0.965 > 0.906 → **no**, go right.
3. Leaf reached: value = -2.720 GPa. Tree 1 adds ν × value = 0.2 × (-2.720) = **-0.544 GPa**.

Tree	Step added (GPa)	Running prediction (GPa)
start (F_0)		16.861
1	-0.544	16.317
2	-0.386	15.930
3	-0.360	15.570
4	-0.217	15.353
5	-0.376	14.977
6 ... 19	-0.285 (total)	...
20	+0.457	15.149

Table 3. The same specimen through all 20 trees. Each tree sends it to one leaf and adds ν × that leaf's value. The final prediction (15.15 GPa) is below the measured 20.83 GPa: this specimen is stiffer than its composition suggests, one of the larger test residuals.

7. Later trees: different splits, smaller corrections

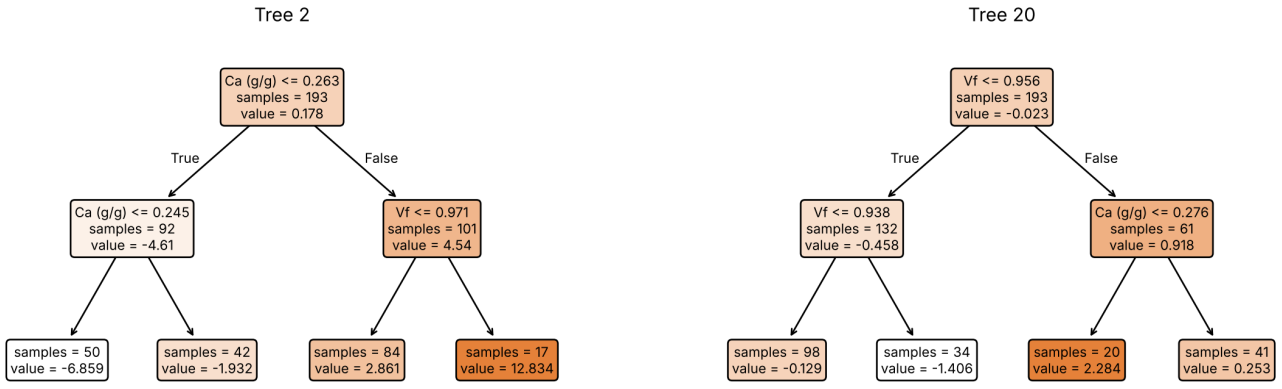


Figure 3. Trees 2 and 20, using the same diagram style (squared error omitted).

Each tree is grown from scratch on a fresh random draw of specimens and on the updated residuals, so its splits can differ. Tree 2 again splits on Ca first (≤ 0.263) but then isolates a small group of 17 very stiff

specimens (Vf > 0.971, leaf value +12.83) that tree 1 left under-predicted. By tree 20 the root split is on Vf, and the largest leaf value is only 2.28 GPa (vs 9.76 for tree 1). Most of the composition trend has been captured and the remaining corrections are small adjustments. This is why the error curve in gbr_boost.ipynb flattens after about 10–15 trees.

In text form (`sklearn.tree.export_text`), tree 1 is:

```
|--- Ca (g/g) <= 0.2640
|   |--- Vf <= 0.9060
|   |   |--- value: [-8.4810]
|   |--- Vf >  0.9060
|   |   |--- value: [-2.7204]
|--- Ca (g/g) >  0.2640
|   |--- Vf <= 0.9561
|   |   |--- value: [1.8624]
|   |--- Vf >  0.9561
|   |   |--- value: [9.7599]
```

Code used to draw and inspect the tree (after fitting `gbr` in the notebook):

```
from sklearn.tree import plot_tree, export_text
tree1 = gbr.estimators_[0, 0]            # a DecisionTreeRegressor
plot_tree(tree1, feature_names=['Ca (g/g)', 'Vf'], filled=True, rounded=True, precision=3)
print(export_text(tree1, feature_names=['Ca (g/g)', 'Vf'], decimals=4))
tree1.tree_.threshold, tree1.tree_.value, tree1.tree_.n_node_samples   # raw arrays
```